

CausalCERTIFAI: A Causality-Aware Framework for Counterfactual Explanations

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This study introduces CausalCERTIFAI, a causality-aware framework for generating counterfactual explanations in tabular machine learning models. The proposed approach integrates graph-based causal discovery with a constraint-aware genetic search process to produce counterfactuals that achieve a desired prediction while remaining consistent with learned causal relationships. Experiments are conducted on multiple classification datasets using decision trees, XGBoost, and LightGBM, comparing standard counterfactual generation with a causal constrained variant. Performance is evaluated in terms of robustness using NCERScore, alongside computational efficiency measured by runtime per instance. Results show that introducing causal constraints tends to increase normalized counterfactual distance, reflecting more restricted but more realistic recourse, with stronger effects observed in higher-dimensional datasets.

Keywords: machine learning, explainable AI, counterfactual explanations, causality, genetic algorithms, model interpretability

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1 Introduction

The growing use of machine learning in high-stakes settings has increased the need for explanation methods to ensure that developed models are transparent and accurate. As model complexity increases, interpretability decreases, making it difficult to explain and justify decisions to stakeholders. This has led to growing interest in explainable artificial intelligence (XAI), particularly in settings where decisions have ethical, legal, or societal implications.

XAI is concerned with techniques that allow people to inspect, anticipate, and place appropriate trust in machine learning systems by making the basis of their predictions interpretable [1]. The field is commonly characterized along several axes: explanations can describe the model as a whole or focus on a single decision, they can rely on a model's internal structure or treat it strictly as an input-output function, and they can either assign importance scores to features or provide example instances that illustrate how the prediction could change [1]. Counterfactual

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explanations are firmly in this last, example-based category: instead of reporting which features mattered, they explain a decision by creating an alternative input that is similar to the original but produces the desired prediction.

A persistent theme in the counterfactual literature is that no single number captures whether a generated counterfactual is of high quality. A method that reliably flips the prediction may still propose changes that are unrealistically large, that touch too many features at once, that recommend edits to attributes a person cannot actually change, or that drift into sparsely populated regions of the feature space. As a result, much of the methodological work concentrates on how counterfactuals are searched for and how feasibility is enforced during the search.

One major family of methods phrases counterfactual generation as a constrained optimization problem in which validity is enforced as either a hard constraint or a penalty, and proximity and sparsity are encouraged via distance-based regularizers. Within this line of work, DCE [11] shows that returning a single minimal counterfactual is often insufficient and instead generates a set of diverse explanations by introducing constraints that prevent previously used feature changes, ensuring that alternative solutions differ in at least one decision variable. Building on this idea, DiCE [2] incorporates diversity directly into the objective function, encouraging the generation of multiple distinct counterfactuals. DACE [6] addresses a complementary weakness of plain distance-based objectives: a candidate that is numerically close to the original instance can still be unrealistic if it lies in a low-density region of the data. By incorporating a distribution-aware penalty into a mixed-integer linear formulation, the method pushes counterfactuals toward parts of the input space that resemble the training distribution, reflecting the idea that changing the original data is not sufficient on its own; a counterfactual must also look like a believable data point.

A second family searches over observed examples instead of optimizing in a continuous space. NICE [3] starts from the nearest unlike neighbor which means the closest training instance that the model assigns to the opposite class, and iteratively copies feature values from this neighbor into the instance being explained, one feature at a time, until the prediction flips. A corresponding limitation is that this strategy depends on having a sufficiently dense neighborhood and on having a meaningful distance for mixed feature types.

A third line of work generates counterfactuals by searching over the model’s decision rules rather than optimizing feature values directly. For tree ensembles, the actionable feature-tweaking approach [4] identifies decision paths that lead to the desired outcome and modifies feature values just enough for the instance to satisfy the conditions along those paths. This way, recourse is obtained by minimally adjusting the original data. For black-box settings, LORE [5] takes a different route: it creates a local neighborhood around an instance using a genetic algorithm, fits an interpretable model, which is a small decision tree, and then extracts explanations.

When the explainer cannot use the model’s internal structure, counterfactual generation becomes a search problem. CADEX [7] iteratively modifies a data instance toward a target class while enforcing constraints such as limiting the number of changed features and maintaining valid values, so that the resulting counterfactual remains realistic. Another black-box, search-based method is CERTIFAI [8], which this study builds on. It uses a custom genetic algorithm to generate counterfactuals under feasibility constraints such as immutability and valid ranges, and since it does not rely on the model’s internal structure, it is a model-agnostic approach.

A common limitation of existing methods is that feasibility is enforced at the individual feature level, while relationships between features are often ignored. Causality-aware approaches address this by ensuring that changes to one variable are propagated to others according to learned dependencies. Methods such as DirectLINGAM [9] discover a causal graph of features and use it to guide counterfactual generation, while OrdCE [10] shows that

Table 1. Comparison of representative counterfactual explanation methods

Method	Reference	Optimization Strategy	Model Agnostic	Data Agnostic	Causality
Feature Tweaking	Tolomei et al. (2017)	Tree-based	No	Only tabular	No
LORE	Guidotti et al. (2018)	Heuristic	Yes	Mainly tabular	No
DCE	Russell et al. (2019)	MI optimization	No	Only tabular	No
CADEX	Moore et al. (2019)	Heuristic	No	Only tabular	Partial
DiCE	Mothilal et al. (2020)	Gradient-based	No	Only tabular	No
DACE	Kanamori et al. (2020)	MI optimization	No	Only tabular	No
CERTIFAI	Sharma et al. (2020)	Heuristic	Yes	Yes	No
OrdCE	Kanamori et al. (2021)	MI optimization	No	Only tabular	Yes
NICE	Brughmans et al. (2024)	Instance-based	Yes	Mainly tabular	No
CausalCERTIFAI		Heuristic	Yes	Only tabular	Yes

the order of changes can also matter. These approaches highlight that realistic counterfactuals require modeling relationships between variables rather than treating them as independent inputs.

A comparison between representative counterfactual explanation methods and the proposed extension is shown in Table 1.

In addition to how counterfactuals are generated, another important aspect is how they are evaluated. This goes beyond simply checking whether a candidate flips the prediction. Robustness in counterfactual explanations is used in two senses. One measures the robustness of the model, asking whether the decision boundary is stable against small perturbations to the input data instance. The other focuses on the robustness of the explanations, asking whether small perturbations to the input lead to similar counterfactual recommendations rather than qualitatively different ones. CERTIFAI [8] adopts the first view, treating the average distance between inputs and their closest valid counterfactuals as a model-robustness score (CERScore): the further away counterfactuals are, the more stable the decision boundary around the original predictions. DiCE-Extended [12] adopts the second view, adding an explicit robustness term to the standard proximity/diversity objective; the term is scored via the Dice-Sørensen coefficient between a generated counterfactual and a perturbed version of it, with weighted loss components trading off proximity, diversity, and robustness. A related idea is stability, which asks a slightly different question: not whether the explanation changes when the input is perturbed, but whether it stays valid when the suggested edit itself is varied slightly. POLARIS [13] captures this by requiring that an explanation hold over a small neighborhood around the proposed change rather than at a single point, and searches for explanations that balance closeness to the original input against the size of the region in which they remain valid. Fairness in counterfactual explanations is often evaluated by comparing the average recourse cost across groups defined by a sensitive attribute: if one group consistently needs larger changes than another to obtain the same outcome, the gap is treated as evidence of unequal access to recourse, an idea operationalized in CERTIFAI [8] through its group-conditional burden measure. Other work argues that fairness can be evaluated at a deeper, causal level. Kusner et al. [14] define *counterfactual fairness* by asking a hypothetical question: would the model have given this individual the same prediction if their protected attribute (such as race or gender) had been different, with everything else that does not depend on it held fixed? If the answer is yes, the prediction is counterfactually fair. This shifts the focus from observed differences between

groups to whether the protected attribute is influencing the prediction at all, even indirectly through other features it shapes. Extending this idea from prediction to recourse, von Kügelgen et al. [15] argue that fair recourse should also account for the downstream effects of the recommended actions, and propose both group- and individual-level fairness criteria for causal recourse. They show that a counterfactually fair predictor does not automatically yield fair recourse, so the two need to be checked separately.

1.1 The Purpose of the Study and Contributions

A key limitation of many counterfactual generation methods is the assumption that features can be modified independently. In real-world data, variables are connected through causal relationships, so a change in one feature typically propagates to others. Ignoring these dependencies can produce counterfactuals that flip the model prediction but violate the underlying structure of the data, recommending actions that are physically or socially infeasible and ultimately reducing trust in the explanation.

This study proposes *CausalCERTIFAI*, an extension of CERTIFAI [8] that incorporates causal constraints into the counterfactual generation process. Instead of treating features independently, the proposed method enforces causal consistency by propagating edits through a causal structure learned from the training data.

The main contribution of this study is the causality-aware extension of the CERTIFAI [8] framework, in which the genetic search is constrained so that perturbing an upstream feature triggers a consistent update of its downstream descendants according to a learned causal graph.

2 The Proposed Method: CausalCERTIFAI

This section presents *CausalCERTIFAI*, a causality-aware framework for generating counterfactual explanations for tabular machine learning models. The framework extends the original CERTIFAI approach, which generates counterfactuals through a genetic search under simple feasibility constraints. Like CERTIFAI, the proposed method is model-agnostic and applies to any predictive model that can be queried for outputs given inputs. It addresses a key limitation of standard counterfactual approaches, including CERTIFAI itself: explanations may achieve the desired prediction but remain unrealistic because they ignore the dependency structure between features.

Given an input instance x and a trained model $f(\cdot)$, the goal is to construct a counterfactual instance x' such that $f(x') = y^*$. In addition to validity, x' should remain close to x , since minimizing the distance places the counterfactual near the decision boundary and reveals which feature changes are sufficient to flip the prediction. The counterfactual must also respect feasibility constraints such as immutable features and valid value ranges. Beyond these standard requirements, *CausalCERTIFAI* enforces causal consistency: edits to actionable features are propagated to their dependents according to a causal structure learned from the data.

2.1 Problem Formulation

Counterfactual generation is formulated as a constrained optimization problem:

$$x' = \arg \min \delta(x, x') \quad \text{s.t.} \quad f(x') = y^*, \quad x' \in \Omega. \quad (1)$$

Here, $\delta(x, x')$ measures the distance between the original and counterfactual instances, capturing the cost of recourse. The constraint $f(x') = y^*$ enforces validity, while Ω defines the feasible set of solutions.

Feasibility is defined at the feature level. Features are partitioned into immutable (I) and actionable (A) sets. Immutable features remain fixed, $x'_I = x_I$, while actionable features may be modified within valid ranges. Departing from the original CERTIFAI formulation, where feature edits are treated independently, the proposed method propagates changes to actionable features through a learned causal structure, ensuring consistency across dependent variables.

2.2 Causal Structure Learning

To model the dependencies between features, CausalCERTIFAI uses DirectLiNGAM [9], a causal discovery method based on linear non-Gaussian structural equation models. Each variable is expressed as a linear function of its direct causes plus an independent noise term, under the assumptions of acyclicity and non-Gaussian noise.

Let $X \in \mathbb{R}^{n \times d}$ denote the data matrix. The algorithm iteratively identifies exogenous variables (variables without parents) by measuring independence between each candidate variable and the residuals obtained from regressing the other variables on it:

$$r_i^{(j)} = x_i - \frac{\text{cov}(x_i, x_j)}{\text{var}(x_j)} x_j. \quad (2)$$

Independence is evaluated using a kernel-based dependence measure $T_{\text{kernel}}(x_j)$, where lower values indicate higher likelihood of exogeneity. After the causal ordering $\mathcal{O}$ has been determined, the causal coefficient matrix C is estimated by regression. Each coefficient C_{ij} represents the direct causal effect of feature j on feature i . Together, $\mathcal{O}$ and C are passed to the counterfactual search and used to enforce causal consistency.

2.3 Counterfactual Search with Causal Propagation

Counterfactuals are generated using a genetic algorithm derived from CERTIFAI and extended with a causal-propagation step. The search begins by initializing a population of candidate solutions through random perturbations of the original instance. The population then evolves over multiple generations using mutation and crossover operators.

Mutation randomly alters selected features, while crossover exchanges feature values between candidate pairs. To enforce causal consistency, feature modifications are propagated through the learned causal structure after each genetic operation. Let Δ denote the vector of feature changes introduced in a candidate $\tilde{x}$. For each feature u taken in causal order $\mathcal{O}$, the propagated update is computed as

$$\Delta_u^{\text{prop}} = \sum_{p: C_{u,p} \neq 0} C_{u,p} \Delta_p, \quad (3)$$

and applied as $\tilde{x}_u \leftarrow \tilde{x}_u + \Delta_u^{\text{prop}}$.

This procedure ensures that downstream variables respond coherently to upstream changes. After propagation, feature bounds are enforced and discrete features are rounded to the nearest integer.

The desired target label y^* is determined automatically from the model prediction. Only candidates satisfying $f(x') = y^*$ are retained, and the closest solutions according to δ are returned.

2.4 Evaluation

A particular model’s robustness is evaluated via counterfactuals, following the formulation introduced in CERTIFAI and computed using the same distance function δ used during the search.

Robustness is quantified using CERScore, the expected distance between each original instance and its corresponding counterfactual:

$$\text{CERScore} = \mathbb{E}[\delta(x, x')]. \quad (4)$$

Higher values in Equation (4) indicate that larger perturbations are required to flip a prediction, and therefore that the model is more robust. To enable comparisons across datasets of different scales, CERScore is normalized by a typical within-class distance:

$$\text{NCERScore} = \frac{\mathbb{E}[\delta(x, x')]}{\sum_{k=1}^K P(y = k) \mathbb{E}[\delta(x_i, x_j) \mid y_i = k, y_j = k]}. \quad (5)$$

Here, the denominator provides a dataset-specific scaling factor that controls for natural variation in feature ranges and class structure. In this study, NCERScore is used as the robustness metric.

2.5 Experimental Pipeline

The experimental pipeline has three stages: data preprocessing and model training, causal discovery, and counterfactual generation with evaluation. First, each dataset is prepared and predictive models are trained. Next, causal relationships are learned using DirectLiNGAM [9], producing a causal ordering $\mathcal{O}$ and a causal effect matrix C . This information is then incorporated into the counterfactual generation process, where CausalCERTIFAI produces feasible and causally consistent explanations. Finally, models are evaluated by using the generated counterfactuals.

2.5.1 Models and Datasets

Three predictive models are used: a decision tree as an interpretable baseline, and two gradient-boosted tree families, XGBoost and LightGBM, representing higher-capacity black-box approaches. The same three models are trained for every dataset.

Evaluation is performed on twelve tabular classification datasets that differ in size, dimensionality, class structure, and missingness patterns: Pima Indians Diabetes, Adult Census Income, Breast Cancer Wisconsin, Iris, Palmer Penguins, Mushroom, Telco Customer Churn, E. coli Protein Localisation, Glass Identification, Dry Bean, Credit Card Default, and Seeds. This range enables a comprehensive assessment of the proposed method across a wide spectrum of settings, from small balanced multi-class problems (Iris, Seeds) to large mixed-type binary ones (Adult, Credit Card Default). Their main characteristics are summarized in Table 2.

2.5.2 Data Preprocessing

All datasets are used in tabular form with minimal preprocessing. Categorical features are encoded into numerical representations to ensure compatibility with the predictive models. Each dataset is split into training and test sets using an 80/20 stratified split, with a fixed random seed of 42 reused across all randomized components to ensure reproducibility.

Some features are count-based quantities (e.g., age or number of events) and must remain discrete. To preserve validity, a discreteness constraint is enforced by rounding such features to the nearest integer after any modification,

Table 2. Summary of datasets used in experiments

Dataset	# Obs.	# Test	# Features	# Classes	Missing (raw)
Pima Indians Diabetes	768	154	8	2	0
Adult Census Income	45,222	9,044	14	2	6,465
Breast Cancer Wisconsin	569	114	30	2	0
Iris	150	30	4	3	0
Palmer Penguins	333	67	6	3	19
Mushroom	8,124	1,625	21	2	2,480
Telco Customer Churn	7,032	1,407	19	2	11
E. coli Protein Localisation	332	67	7	6	0
Glass Identification	214	43	9	6	0
Dry Bean	13,611	2,723	16	7	0
Credit Card Default	30,000	6,000	23	2	0
Seeds	210	42	7	3	0

including causal propagation. For instance, in the Pima Indians Diabetes dataset, *Pregnancies* and *Age* are treated as discrete to avoid implausible fractional values.

2.5.3 Model Training

Model hyperparameters are tuned by grid search with 5-fold cross-validation on the training data. The configuration with the highest cross-validated accuracy is selected, the model is retrained on the full training split, and accuracy is reported on the test set as the primary performance metric.

2.5.4 Counterfactual Generation

Counterfactual explanations are generated using the genetic search procedure introduced by CERTIFAI. Before generation, causal relationships are learned from the reference dataset X using DirectLiNGAM, yielding the causal ordering $\mathcal{O}$ and coefficient matrix C .

Counterfactuals are generated for test instances $x_i \in D$. For each model, two settings are compared: with causal constraints enabled and without them. The main search parameters are set as follows: mutation probability $P_m = 0.7$, crossover probability $P_c = 0.5$, number of generations $G = 10$, population size $k_s = 1000$, generation retain size $k_g = 500$, and final number of counterfactuals per instance $k_f = 1$. The distance metric is $\delta = L_1$. To keep the counterfactual search tractable on the larger datasets, it is run on the first 150 instances of the test set when the test set has at least 150 rows; smaller test sets are used in their entirety.

To ensure feasibility, feature-wise bounds are enforced throughout the search:

$$x'_j \in [\min(X_j), \max(X_j)]. \quad (6)$$

When causal constraints are enabled, feature modifications are propagated through the causal structure using C and $\mathcal{O}$, which ensures consistency between dependent variables. After propagation, bounds and discreteness constraints are re-applied.

3 Results

This section reports the empirical results. For each dataset, three predictive models (decision tree, XGBoost, and LightGBM) are trained, and counterfactual explanations are generated under two settings: with causal constraints and without causal constraints. Results are summarized using predictive accuracy, counterfactual robustness (NCERScore), and runtime per generated counterfactual.

3.1 Predictive Performance and Counterfactual Robustness

Table 3 presents predictive accuracy together with NCERScore for both CERTIFAI and the proposed CausalCERTIFAI settings. Accuracy reflects standard predictive performance on the test split, while NCERScore measures the average distance required to reach a valid counterfactual. Higher NCERScore values indicate that larger changes are needed to alter the prediction, which corresponds to greater robustness. In Table 3, for each dataset, the bold value in each counterfactual setting marks the most robust model, i.e. the one with the highest NCERScore.

Across datasets, gradient-boosted models generally achieve higher predictive accuracy than the decision tree, most visibly on the E. coli dataset. Several datasets (e.g., Iris, Mushroom, Penguins) are perfectly classified by all three models.

NCERScore values vary substantially across datasets. On Breast Cancer, Dry Bean, Glass, Mushroom, and Seeds, enabling causality increases NCERScore for all three models, indicating that causal constraints require larger changes to reach valid counterfactuals. The effect is particularly pronounced on Breast Cancer. In contrast, on Adult, Iris, Credit Card Default, Pima Diabetes and E. coli, the effect is mixed, with some models showing lower NCERScore under causality. On simpler datasets such as Penguins and Telco Churn, differences between causal and non-causal settings are small. However, causal and non-causal NCERScore values are not strictly comparable: they are produced by different optimization problems with different feasibility constraints. A higher NCERScore under causality does not mean the model has become more robust, since the model itself is unchanged. What changes is the set of admissible perturbations, and hence the search space of the problem.

3.2 Runtime Analysis

Across all datasets, causal counterfactual generation consistently requires more time per instance than the non-causal setting, with slowdowns ranging from modest on the smallest datasets to roughly 2–3 $\times$ on the larger ones. Two factors appear to be the main drivers of the increased time complexity. The first is feature dimensionality, most clearly visible on Breast Cancer (30 numeric features, ≈ 60 s per counterfactual under causality), where the per-feature cost of causal propagation leads to a substantial amount of time overall. The second is data type: on Mushroom, where all 21 features are categorical, the discreteness constraint is enforced after every mutation and propagation step, which substantially raises the per-counterfactual cost (≈ 20 s) even though Mushroom has fewer features than Breast Cancer. Credit Card Default (23 mixed-type features, ≈ 25 s) sits between the two cases.

On smaller datasets (e.g., Iris, E. coli, Penguins, Seeds), runtime differences remain modest (≤ 0.5 s per counterfactual), suggesting that the cost of enforcing causal consistency depends jointly on feature dimensionality and the number of categorical or integer-valued features, since these add discreteness constraints that complicate the genetic search.

Table 3. Accuracy and NCERScore for CausalCERTIFAI and CERTIFAI

Dataset	Model	Accuracy	NCERS (CausalCERTIFAI)	NCERS (CERTIFAI)
Adult	DT	0.85	0.17	0.17
	XGB	0.87	0.17	0.19
	LGBM	0.87	0.17	0.17
Breast Cancer	DT	0.95	0.54	0.41
	XGB	0.97	0.51	0.11
	LGBM	0.96	0.51	0.27
Credit Card Default	DT	0.82	4.69	4.69
	XGB	0.82	5.01	4.95
	LGBM	0.82	4.86	4.91
Dry Bean	DT	0.91	0.48	0.30
	XGB	0.93	0.44	0.29
	LGBM	0.93	0.48	0.28
E. coli	DT	0.84	2.77	2.73
	XGB	0.91	2.72	2.82
	LGBM	0.90	2.73	2.67
Glass	DT	0.77	1.66	1.57
	XGB	0.74	1.81	1.74
	LGBM	0.79	1.70	1.60
Iris	DT	1.00	0.97	1.03
	XGB	1.00	1.02	0.87
	LGBM	1.00	1.14	0.99
Mushroom	DT	1.00	0.66	0.62
	XGB	1.00	0.66	0.62
	LGBM	1.00	0.66	0.61
Penguins	DT	1.00	0.06	0.07
	XGB	1.00	0.07	0.05
	LGBM	1.00	0.06	0.06
Pima Diabetes	DT	0.80	0.63	0.64
	XGB	0.79	0.54	0.53
	LGBM	0.79	0.55	0.53
Seeds	DT	0.93	0.88	0.74
	XGB	0.93	0.91	0.84
	LGBM	0.93	0.95	0.82
Telco Churn	DT	0.79	0.02	0.02
	XGB	0.79	0.03	0.02
	LGBM	0.80	0.02	0.02

The additional computational cost introduced by causal constraints is an important consideration, particularly when generating many counterfactuals. The effect is most pronounced on larger or more complex datasets. According to Table 4, on Breast Cancer, the runtime increases from roughly 19 seconds to around 60 seconds per counterfactual, and on Credit Card Default from about 9–10 seconds to around 25 seconds. Similarly, on Mushroom, the cost rises from about 7 seconds to approximately 20 seconds.

4 Discussion

The empirical results show that adding causal structure to the CERTIFAI framework changes the counterfactuals it produces in meaningful ways. Across twelve datasets, the causality-aware version generally generates measurably different counterfactuals than the unconstrained search, with consistent patterns. On Breast Cancer, Dry Bean, Glass, Mushroom, and Seeds, all three models needed larger feature changes to flip their predictions once causal propagation was enforced. Breast Cancer showed the largest gap between the two settings. This matches the motivation behind CausalCERTIFAI: when downstream features can be edited independently of their causal parents, the search finds counterfactuals with small raw distances but unrealistic feature combinations. Enforcing causal propagation removes these and pushes the search toward more realistic recourse.

The biggest limitation of the proposed method is the loss of data-agnosticism. The original CERTIFAI works on any input type: its black-box genetic search runs on tabular data, images, and text. CausalCERTIFAI does not. The propagation step needs a causal graph over named features, which only makes sense for tabular data. The method stays general within tabular settings, but it no longer works across data types.

A second limitation is computational cost. Enabling the causal setting increases the per-counterfactual runtime on every dataset. On small datasets the increase is minor, on larger ones it reaches two to three times. On low-dimensional datasets (Iris, E. coli, Penguins, Seeds), the change is negligible. On higher-dimensional datasets, the cost grows quickly: Breast Cancer’s thirty continuous features push the per-counterfactual runtime to about sixty seconds under causality, because propagation has to update every downstream variable after each genetic operation, and this cost accumulates over ten generations with a population of one thousand. A different kind of cost appears on datasets dominated by categorical features. Mushroom is the clearest example: with all twenty-one features categorical and label-encoded, per-counterfactual runtime reaches twenty seconds, which is more than feature count alone would suggest. The likely reason is that the discreteness constraint has to be enforced after every mutation and propagation step.

In summary, this study contributes a model-agnostic, search-based counterfactual generator that incorporates causal structure into its feasibility constraints, along with an empirical comparison between the causal and non-causal settings across twelve tabular datasets. The main result is that the causality-aware version produces more realistic counterfactuals by removing edits that violate dependencies between features, at the cost of a runtime increase that reaches roughly $2\text{--}3\times$ on high-dimensional or heavily categorical data. The most useful direction for future work is restoring data-agnosticism so that the method can be applied to non-tabular inputs such as images or text. Another useful direction is designing genetic operators and causality perturbations more efficiently to bring down the runtime.

Table 4. Time complexity per generated counterfactual (seconds) for CausalCERTIFAI versus CERTIFAI

Dataset	Model	$t_{\text{CausalCERTIFAI}}$	t_{CERTIFAI}
Adult	DT	2.54	1.44
	XGB	2.83	1.66
	LGBM	2.71	1.56
Breast Cancer	DT	59.92	18.66
	XGB	60.26	19.12
	LGBM	60.20	18.82
Credit Card Default	DT	24.79	9.47
	XGB	25.37	9.58
	LGBM	26.44	10.09
Dry Bean	DT	8.30	3.22
	XGB	7.94	3.46
	LGBM	7.77	3.33
E. coli	DT	0.41	0.33
	XGB	0.52	0.42
	LGBM	0.47	0.39
Glass	DT	1.02	0.65
	XGB	1.23	0.82
	LGBM	1.15	0.72
Iris	DT	0.12	0.09
	XGB	0.14	0.11
	LGBM	0.13	0.10
Mushroom	DT	19.85	6.95
	XGB	20.15	7.21
	LGBM	20.08	7.08
Penguins	DT	0.30	0.22
	XGB	0.37	0.28
	LGBM	0.35	0.27
Pima Diabetes	DT	0.68	0.48
	XGB	0.86	0.63
	LGBM	0.81	0.56
Seeds	DT	0.51	0.32
	XGB	0.67	0.41
	LGBM	0.59	0.37
Telco Churn	DT	11.25	5.08
	XGB	11.65	5.33
	LGBM	11.28	5.92

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